

Detailed Model Analysis for Betel Leaf Disease Classification Using Hybrid CNN–Swin Transformer and Attention Mechanisms

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¹2022-1-60-122 ²2022-2-60-124 ³2022-1-60-362 ⁴2022-3-60-080

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Abstract—The economic sustainability of betel leaf (*Piper betle* L.) cultivation is severely threatened by rapid disease proliferation, often causing significant yield loss and financial instability for farmers. Traditional manual diagnosis is labor-intensive, subjective, and prone to error, necessitating the development of automated, precision-driven diagnostic tools. This study proposes a novel Custom Hybrid CNN–Swin Transformer architecture that synergizes the local feature extraction capabilities of Convolutional Neural Networks (CNNs) with the global context modeling of Swin Transformers to accurately classify betel leaf diseases. Twelve distinct deep-learning frameworks are rigorously evaluated, including industry-standard baselines (InceptionV3, ResNet50), custom lightweight CNNs, and attention-enhanced models (CBAM-CNN, SE-ResNet) using a balanced dataset of 1,000 images and a secondary imbalanced dataset for robustness assessment. The proposed Hybrid model achieves a state-of-the-art accuracy of 98% on the primary dataset and demonstrates superior generalization with 93% accuracy on the imbalanced set, outperforming all standard benchmarks. Explainable AI (XAI) via Grad-CAM is integrated to visualize decision boundaries, confirming the model’s clinical reliability by focusing on relevant pathological lesions rather than background noise. These findings underscore the potential of hybrid attention-based architectures in precision agriculture.

Index Terms—Betel Leaf Disease, Hybrid CNN–Swin Transformer, Deep Learning, Explainable AI (XAI), Precision Agriculture, Attention Mechanisms, CBAM, Grad-CAM

I. INTRODUCTION

Betel leaf (*Piper betle* L.), commonly known as “Paan,” is a cash crop of immense medicinal and economic significance across South and Southeast Asia. It provides livelihoods for millions of farmers and holds a prominent role in cultural and traditional practices. However, the crop is highly susceptible to devastating diseases including Bacterial Leaf Spot, Foot Rot, and Fungal Brown Spot, which can cause total crop failure if not detected in early stages [1].

The current reliance on manual visual inspection is fraught with limitations: it requires expert knowledge, is time-consuming, and is often inconsistent due to human fatigue or symptom similarity. As climate change accelerates pathogen proliferation, there is an urgent need for intelligent, automated diagnostic systems capable of rapid and accurate field assessment.

Recent advances in Computer Vision and Deep Learning (DL) have revolutionized precision agriculture, enabling

automated plant disease detection with remarkable success. CNNs have become the standard for such tasks, excelling at extracting local textural features from leaf images. Models such as ResNet, Inception, and MobileNet have been widely adopted for various crop pathologies [2]. However, standard CNNs often struggle to capture long-range dependencies due to their fixed receptive fields, limiting performance when distinguishing subtly or widely distributed disease patterns [3]. Furthermore, many existing studies focus on balanced datasets, failing to address class imbalance common in real-world agricultural scenarios, and often lack model interpretability—a critical barrier to end-user trust.

To address these challenges, this study introduces a comprehensive comparative analysis of fourteen deep learning architectures, proposing a Custom Hybrid CNN–Swin Transformer as a robust solution. The primary contributions are as follows:

- 1) **Novel Hybrid Architecture:** A Hybrid CNN–Swin Transformer that outperforms traditional CNN benchmarks by integrating local and global feature extraction.
- 2) **Comprehensive Benchmarking:** Evaluation of 14 diverse models ranging from lightweight custom CNNs to heavy transfer-learning baselines (InceptionV3, EfficientNet-B0).
- 3) **Robustness Analysis:** Validation on a secondary, highly imbalanced dataset to ensure real-world applicability.
- 4) **Explainability Integration:** Grad-CAM visualization to confirm that classifications are based on relevant disease symptoms rather than artifacts, enhancing clinical trust.

II. RELATED WORK

A. Classical Machine Learning Approaches

Early automated diagnosis relied on hand-crafted feature extraction followed by classical classifiers. Studies utilizing Support Vector Machines (SVM), Random Forests (RF), and K-Nearest Neighbors (KNN) based on color, texture, and shape descriptors established baselines, often achieving accuracies between 63% and 85% [15]. These methods are limited by dependence on manual feature engineering and poor generalization to unstructured environments.

B. Deep Learning and Transfer Learning

The advent of CNNs marked a paradigm shift. Transfer learning, where models pre-trained on ImageNet are fine-tuned for specific tasks, became the dominant strategy. Architectures such as DenseNet201 and EfficientNet-B0 have reported accuracies exceeding 99% on specific datasets [6]. However, heavy architectures incur high computational costs, making field deployment on resource-constrained mobile devices challenging. Lightweight models like MobileNetV3-Small have shown promise for real-time applications, though sometimes at the cost of reduced feature granularity [16].

C. Hybrid and Attention-Based Mechanisms

To overcome the limitations of standard CNNs in capturing global context, recent literature has explored hybrid architectures and attention mechanisms. The combination of CNNs with Transformer models has gained traction for modeling long-range dependencies. Studies integrating Swin Transformer blocks with YOLOv5 backbones have shown significant improvements in detection precision (mAP) compared to baseline detectors [13]. Semi-supervised learning frameworks have also been proposed for limited-label scenarios, achieving high performance but often requiring complex training pipelines [6].

D. Research Gaps

Despite these advancements, several critical gaps remain. A recurring limitation is the reliance on small or region-specific datasets with limited class diversity, hampering model generalizability [11]. Furthermore, few studies rigorously test robustness against class imbalance or validate model decisions using Explainable AI (XAI) techniques. This study addresses these voids by proposing a robust Hybrid CNN–Swin Transformer validated on both balanced and imbalanced datasets, and substantiated with Grad-CAM visual explanations.

III. METHODOLOGY

A. Dataset Description

Two datasets from open-source repositories were used to validate model robustness and generalizability.

Dataset 1 (Balanced): The primary dataset consists of 1,000 images distributed evenly across four classes: Bacterial Leaf Disease, Dried Leaf, Fungal Brown Spot Disease, and Healthy Leaf (250 images per class). This balanced distribution ensures unbiased training and evaluation [17].

Dataset 2 (Imbalanced): A secondary dataset characterized by significant class imbalance—Healthy Leaf (1,080 images), Leaf Rot (269 images), and Leaf Spot (688 images)—was used to evaluate model robustness under real-world conditions.

B. Data Preprocessing and Augmentation

A rigorous preprocessing pipeline was applied:

- **Resizing:** All images were resized to 224×224 pixels (299×299 for InceptionV3).
- **Normalization:** Pixel values normalized using ImageNet mean and standard deviation.

- **Augmentation:** Random rotations, horizontal flips, and color jitter were applied on-the-fly during training.
- **Splitting:** Dataset 1 was split into Training (70%), Validation (15%), and Test (15%) using stratified sampling.

C. Training Configuration

All models were trained on a CUDA-enabled GPU (Tesla P100/T4) using the Adam optimizer with a dynamic learning rate and Cross-Entropy Loss. Early Stopping (patience = 5–10 epochs) was implemented to prevent overfitting and ensure computational efficiency.

D. Model Architectures

1) *Category 1 – Transfer Learning Baselines:* Industry-standard models pre-trained on ImageNet served as benchmarks: **VGG16**, **ResNet-50**, **InceptionV3**, **MobileNetV2**, **DenseNet121**, **EfficientNet-B0**, **Xception**, and **AlexNet**.

2) *Category 2 – Custom Architectures:*

- **Custom Sequential CNN:** Standard baseline built with Conv2D $\rightarrow$ ReLU $\rightarrow$ MaxPool layers, trained from scratch for 35 epochs.
- **BLCNN (Betel Leaf CNN):** A domain-specific architecture utilizing depth-wise separable convolutions to reduce parameters while maintaining accuracy [3].

3) *Category 3 – Advanced Attention & Hybrid Models:*

- **SE-Custom CNN:** Custom CNN augmented with Squeeze-and-Excitation (SE) blocks for channel-wise feature recalibration [4].
- **SE-ResNet:** ResNet backbone enhanced with SE blocks.
- **CBAM-CNN:** Convolutional Block Attention Module applying both channel and spatial attention—“what to focus on” and “where to look.”
- **Hybrid CNN–Swin Transformer:** The proposed architecture, fusing a CNN stem for local feature extraction with Swin Transformer blocks for long-range global context understanding.

E. Proposed Architecture: Hybrid CNN–Swin Transformer

The proposed model consists of three key components:

- 1) **CNN Stem:** Two Conv–BN–GELU layers reduce the 224×224 input to a 56×56 feature map, capturing low-level textural features (edges, shapes).
- 2) **Swin Transformer Blocks:** Stages 1 and 2 of Swin-Tiny capture long-range dependencies and global context using shifted-window self-attention, enabling the model to analyze holistic leaf structure.
- 3) **Classification Head:** A post-processing Conv layer followed by Global Average Pooling (GAP) and a Dropout $\rightarrow$ Linear $\rightarrow$ ReLU $\rightarrow$ Linear classifier head produces the final 4-class prediction.

This synergistic design allows the model to simultaneously capture fine lesion-level details (CNN) and global leaf structure context (Transformer), which is critical for discriminating between visually similar disease patterns.

F. Explainability: Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to visualize which regions of each leaf image drove the model’s prediction. This “opens the black box,” confirming that top-performing models focus on pathologically relevant regions (lesions, necrotic spots) rather than background artifacts.

IV. RESULTS AND ANALYSIS

A. Comparative Performance Overview

Table I summarizes the test accuracy, training runtime, and dataset for all 14 architectures evaluated in this study.

TABLE I
COMPARATIVE PERFORMANCE OF ALL MODELS

Rank	Model	Acc.	Runtime	Dataset
1	Hybrid CNN–Swin Transformer	98.00%	8m 31s	D1
2	CBAM-CNN	97.33%	24.98s	D1
3	InceptionV3	97.00%	3m 09s	D1
4	SE-Custom CNN	95.00%	6m 47s	D1
5	ResNet-50	95.00%	1m 56s	D1
6	MobileNetV2	94.00%	1m 44s	D1
7	DenseNet121	94.00%	2m 19s	D1
8	Hybrid CNN–Swin (D2)	93.00%	4m 19s	D2
9	AlexNet	92.00%	49.96s	D1
10	VGG16	91.00%	2m 19s	D1
11	Custom CNN	90.00%	20.66s	D1
12	Xception	89.00%	4m 02s	D1
13	BLCNN	88.87%	31.62s	D1
14	SE-ResNet	88.00%	7m 26s	D1
15	EfficientNet-B0	87.00%	1m 16s	D1

D1 = Dataset 1 (Balanced); D2 = Dataset 2 (Imbalanced)

B. Transfer Learning Baselines

InceptionV3 emerged as the top-performing standard CNN with 97.0% accuracy, achieved in only 3.09 minutes. Its multi-scale Inception modules effectively capture disease spots of varying sizes, demonstrating excellent generalization [2], [5].

ResNet-50 achieved 95.0% with the fastest convergence among heavy models (1.56 min), validating the effectiveness of skip connections for solving vanishing gradient issues on this task.

MobileNetV2 and **DenseNet121** both reached 94.0%. MobileNetV2’s depth-wise separable convolutions make it an ideal candidate for mobile deployment. DenseNet121’s feature reuse incurred higher computational cost without commensurate accuracy gain.

A notable finding was that **AlexNet** (92.0%) outperformed more modern architectures including **Xception** (89.0%) and **EfficientNet-B0** (87.0%). This suggests that for smaller, specialized datasets with distinct morphological features, simpler inductive biases—such as AlexNet’s large filters—can be more effective than the complex compound scaling strategies designed for massive datasets. EfficientNet-B0 likely required more extensive hyperparameter tuning to leverage its scaling capabilities.

C. Custom Architectures

The **Custom Sequential CNN** achieved a competitive 90.0% accuracy trained from scratch in only 20 seconds. While it demonstrates a strong baseline, it lacked the capacity to capture fine-grained textures distinguishing bacterial versus fungal spots.

The **BLCNN**, though domain-specific, performed slightly below the standard Custom CNN (88.87%), indicating that architectural tweaks from literature did not fully transfer to our dataset split.

D. Attention-Enhanced Models

CBAM-CNN achieved the second-highest accuracy of 97.33% in only 25 seconds of training—a remarkable efficiency-to-performance ratio. Spatial attention enabled the model to suppress background noise and focus exclusively on leaf lesions, outperforming all pure CNN baselines including InceptionV3.

SE-Custom CNN achieved 95.0%, a significant 5% improvement over the base Custom CNN (90.0%), demonstrating that channel-wise attention substantially boosts feature extraction in lightweight models.

Conversely, **SE-ResNet** underperformed at 88.0%, indicating that simply augmenting a heavy backbone with attention blocks does not guarantee improvement without careful architecture-specific tuning.

E. Proposed Hybrid CNN–Swin Transformer

The Hybrid CNN–Swin Transformer achieved the highest overall accuracy of **98.0%** on Dataset 1. The CNN stem captures low-level edge features while the Swin Transformer blocks model the holistic leaf structure through self-attention over shifted windows. This synergy enables the model to simultaneously identify localized disease lesions and global structural anomalies.

Critically, when evaluated on the imbalanced Dataset 2, the model maintained **93.0%** accuracy, demonstrating robust generalization under real-world data conditions—a property absent in many prior works restricted to balanced datasets.

F. Explainability Analysis (Grad-CAM)

Grad-CAM heatmaps confirmed that the Hybrid Swin and CBAM-CNN models correctly focused on bacterial lesions and necrotic spots rather than spurious background features. Weaker models tended to distribute attention diffusely across the image. The ROC curves for the top models displayed Area Under the Curve (AUC) scores approaching 1.0 across all four disease classes, confirming exceptional discriminatory power.

V. DISCUSSION

A. Architecture Insights

The results reveal several important architectural insights for small-scale, domain-specific classification tasks:

- **Multi-scale processing is valuable:** InceptionV3’s multi-scale Inception modules effectively detected spots of

varying sizes, explaining its dominance among standard CNNs.

- **Complexity does not guarantee accuracy:** AlexNet outperformed EfficientNet-B0, suggesting that simpler inductive biases may suit small specialized datasets better than compound scaling.
- **Attention mechanisms provide consistent improvement:** CBAM and SE blocks reliably improved over base architectures by directing model focus to disease-relevant regions.
- **Hybrid local-global modeling is optimal:** The CNN–Swin fusion achieved the best balance of accuracy and generalization, validating that leaf disease diagnosis benefits from both local texture capture (CNN) and global structure awareness (Transformer).

B. Practical Deployment Considerations

For field deployment, two recommendations emerge from this study:

- 1) **High-accuracy server-side deployment:** The Hybrid CNN–Swin Transformer is recommended for cloud-based or server-side diagnosis due to its 98% accuracy and robust generalization.
- 2) **Mobile/edge deployment:** CBAM-CNN (97.33% accuracy, 25-second training, lightweight design) or MobileNetV2 (94%) are recommended for resource-constrained mobile applications.

VI. CONCLUSION

This research successfully completed a theoretical and experimental evaluation of automated betel leaf disease detection across fourteen different deep learning frameworks. The study demonstrates that while standard pre-trained CNNs like InceptionV3 offer high accuracy, the integration of attention mechanisms and hybrid architectures yields superior results.

The Custom Hybrid CNN–Swin Transformer emerged as the state-of-the-art model for this task, achieving a benchmark accuracy of 98% on the primary dataset and 93% on the imbalanced dataset. This confirms that the synergy between local convolutional feature extraction and the global attention mechanisms of Swin Transformers provides the most comprehensive approach for identifying subtle and widespread leaf pathologies. The integration of Grad-CAM further ensures clinical interpretability—an essential requirement for real-world agricultural adoption.

Future work will focus on: (1) model quantization for edge device deployment; (2) semantic segmentation using U-Net architectures to measure infection severity; (3) dataset expansion to include varying lighting conditions and additional disease types; and (4) exploration of Vision Transformers (ViT) for comparison with the CNN-based models evaluated in this study.

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